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Inventor(s)

HAN; Jonghyun et al.

ELECTRONIC DEVICE FOR RECOMMENDING INFORMATION ACCORDING TO SITUATION AND METHOD FOR CONTROLLING THE SAME

Abstract

An electronic device is disclosed. An electronic device according to the disclosure may include a display, a memory, and at least one processor, wherein the at least one processor is configured to execute instructions to, based on a content displayed on the display satisfying a configured condition, acquire information included in the content, arrange, based on a first priority, plurality of items acquired based on the information included in the content and store in the memory, rearrange the plurality of items, based on a second priority based on a current time point satisfying the configured condition, and control the display to display the plurality of rearranged items.

Inventors: HAN; Jonghyun (Suwon-si, KR), KIM; Sangheon (Suwon-si, KR), KIM; Seokwon (Suwon-si, KR), KIM; Sumi (Suwon-si, KR), KIM; Hyunmin (Suwon-si, KR), LEE; Seungyong (Suwon-si, KR), JEON; Kicheol (Suwon-si, KR), HONG; Seungpyo (Suwon-si, KR), OH; Younghak (Suwon-si, KR)

Applicant: SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/097,025, filed Jan. 13, 2023, which is a continuation of International Application No. PCT/KR2023/000049, filed Jan. 2, 2023, designating the United States, in the Korean Intellectual Property Receiving Office, and claiming priority to KR Patent Application No. 10-2022-0030674, filed Mar. 11, 2022, and to KR Patent Application No. 10-2022-0006212, filed Jan. 14, 2022, the disclosures of which are all hereby incorporated by reference herein in their entireties.

BACKGROUND

Field

[0002] Embodiments of the disclosure relate to an electronic device for recommending information according to a situation, and a method for controlling the same.

Description of Related Art

[0003] Various services and additional functions provided through electronic devices, for example, a portable electronic device such as a smartphone, are gradually increasing. In order to increase the utility value of such electronic devices and satisfy the needs of various users, communication service providers or electronic device manufacturers offer various functions and develop electronic devices competitively to differentiate them from other companies. Accordingly, various functions provided through electronic devices are becoming more advanced.

[0004] Electronic devices provide various graphic user interfaces (GUIs) for interaction with users through a display.

[0005] In order for a user to acquire information, an electronic device recommends related information, based on information input by the user, so as to improve a situation that the user needs to directly search for and record desired information at a desired time.

[0006] In this way, when related information is recommended based on the information pre-input by the user, information that the user is not actually interested in may be recommended in some cases, and information may be recommended at a time point irrelevant to a time point at which the user needs the recommendation information.

SUMMARY

[0007] Information related to information that is not input by the user may not be provided to the user even though the information is desired by the user.

[0008] Various embodiments of the disclosure may provide an electronic device for providing information desired by a user in a timely manner according to a situation, and a method for controlling the same.

[0009] According to an embodiment, an electronic device may include a display, a memory, and at least one processor, wherein the at least one processor is configured to execute instructions to, if a content displayed on the display satisfies a configured condition, acquire information included in

the content, arrange, based on a first priority, multiple items acquired based on the information included in the content so as to store the same in the memory, rearrange the multiple items, based on a second priority if a current time point satisfies the configured condition, and control the display to display the rearranged multiple items.

[0010] According to an embodiment, a control method of an electronic device may include, if a content displayed on a display of the electronic device satisfies a configured condition, acquiring information included in the content, arranging, based on a first priority, multiple items acquired based on the information included in the content so as to store the same in a memory of the electronic device, rearranging the multiple items, based on a second priority if a current time point satisfies the configured condition, and controlling the display to display the rearranged multiple items.

[0011] An electronic device, according to an embodiment of the disclosure, may provide a user with desired information in a timely manner by changing a priority of information in connection with the user's schedule.

[0012] The electronic device, according to an embodiment of the disclosure, may acquire information based on a user's interest and recommends information based thereon so as to provide more appropriate information to a user.

[0013] The electronic device, according to an embodiment of the disclosure, may provide updated information to a user in real time by rearranging the updated information with the highest priority.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is a block diagram of an electronic device in a network environment according to an embodiment;

[0015] FIG. 2 is a flowchart for illustrating an operation of an electronic device for rearranging multiple items, based on a current time point according to an embodiment;

[0016] FIG. 3 is a flowchart for illustrating an operation of an electronic device for rearranging multiple items, based on a change of information according to an embodiment;

[0017] FIG. 4A is a diagram for illustrating an operation of acquiring information included in a content displayed on a display of an electronic device, according to an embodiment;

[0018] FIG. 4B is a diagram for illustrating an operation of arranging multiple items acquired based on information included in a content of an electronic device according to a priority and displaying the arranged multiple items, according to an embodiment;

[0019] FIG. 4C is a diagram for illustrating an operation of rearranging, based on a changed priority, the arranged multiple items and displaying the rearranged multiple items, according to an embodiment;

[0020] FIG. 5A is a diagram for illustrating an operation of acquiring information included in a content displayed on a display of an electronic device, according to an embodiment;

[0021] FIG. 5B is a diagram for illustrating an operation of acquiring information included in a content displayed on the display of the electronic device, according to an embodiment;

[0022] FIG. 5C is a diagram for illustrating an operation of acquiring information included in a content displayed on the display of the electronic device, according to an embodiment;

[0023] FIG. 5D is a diagram for illustrating an operation of acquiring information included in a content displayed on the display of the electronic device, according to an embodiment;

[0024] FIG. 5E is a diagram for illustrating an operation of acquiring information included in a content displayed on the display of the electronic device, according to an embodiment;

[0025] FIG. 6 is a diagram for illustrating an operation of acquiring information included in a content displayed on a display of an electronic device, according to an embodiment;

[0026] FIG. 7A is a diagram for illustrating an operation of an electronic device when multiple items are rearranged based on changed priorities, according to an embodiment;

[0027] FIG. 7B is a diagram for illustrating an operation of the electronic device when multiple items are rearranged based on changed priorities, according to an embodiment;

[0028] FIG. 7C is a diagram for illustrating an operation of the electronic device when multiple items are rearranged based on changed priorities, according to an embodiment;

[0029] FIG. 7D is a diagram for illustrating an operation of the electronic device when multiple items are rearranged based on changed priorities, according to an embodiment;

[0030] FIG. 8A is a diagram for illustrating electronic device operations of acquiring information, arranging multiple items based on the acquired information according to priorities, and displaying the arranged multiple items, according to an embodiment;

[0031] FIG. 8B is a diagram for illustrating electronic device operations of rearranging, based on changed priorities, the multiple items and displaying the rearranged multiple items, according to an embodiment;

[0032] FIG. 8C is a diagram for illustrating electronic device operations of rearranging, based on changed priorities, the multiple items and displaying the rearranged multiple items, according to an embodiment;

[0033] FIG. 8D is a diagram for illustrating electronic device operations of rearranging, based on priorities, the multiple items and displaying the rearranged multiple items, according to an embodiment;

[0034] FIG. 8E is a diagram for illustrating electronic device operations of rearranging, based on priorities, the multiple items and displaying the rearranged multiple items, according to an embodiment;

[0035] FIG. 8F is a diagram for illustrating electronic device operations of rearranging, based on priorities, the multiple items and displaying the rearranged multiple items, according to an embodiment;

[0036] FIG. 8G is a diagram for illustrating electronic device operations of rearranging, based on priorities, the multiple items and displaying the rearranged multiple items, according to an embodiment;

[0037] FIG. 9A is a diagram for illustrating electronic device operations of acquiring information, arranging multiple items based on the acquired information according to priorities, and displaying the arranged multiple items, according to an embodiment;

[0038] FIG. 9B is a diagram for illustrating electronic device operations of rearranging, based on priorities, the multiple items and displaying the rearranged multiple items, according to an embodiment;

[0039] FIG. 9C is a diagram for illustrating electronic device operations of rearranging, based on priorities, the multiple items and displaying the rearranged multiple items, according to an embodiment;

[0040] FIG. 9D is a diagram for illustrating electronic device operations of rearranging, based on priorities, the multiple items and displaying the rearranged multiple items, according to an embodiment;

[0041] FIG. 10A is a diagram for illustrating electronic device operations of acquiring information, arranging multiple items based on the acquired information according to priorities, and displaying the arranged multiple items, according to an embodiment;

[0042] FIG. 10B is a diagram for illustrating electronic device operations of rearranging, based on priorities, the multiple items and displaying the rearranged multiple items, according to an embodiment;

[0043] FIG. 10C is a diagram for illustrating electronic device operations of rearranging, based on priorities, the multiple items and displaying the rearranged multiple items, according to an embodiment;

[0044] FIG. **10D** is a diagram for illustrating electronic device operations of rearranging, based on priorities, the multiple items and displaying the rearranged multiple items, according to an embodiment;

[0045] FIG. **11A** is a diagram for illustrating electronic device operations of acquiring information, arranging multiple items based on the acquired information according to priorities, and displaying the arranged multiple items, according to an embodiment;

[0046] FIG. **11B** is a diagram for illustrating electronic device operations of rearranging, based on priorities, the multiple items and displaying the rearranged multiple items, according to an embodiment; and

[0047] FIG. **11C** is a diagram for illustrating electronic device operations of rearranging, based on priorities, the multiple items and displaying the rearranged multiple items, according to an embodiment.

DETAILED DESCRIPTION

[0048] FIG. **1** is a block diagram illustrating an electronic device **101** in a network environment **100** according to various embodiments. Referring to FIG. **1**, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

[0049] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0050] The auxiliary processor **123** may control, for example, at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active (e.g., executing an

application) state. According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence model is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0051] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0052] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0053] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0054] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0055] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0056] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or an external electronic device (e.g., an electronic device **102** (e.g., a speaker or a headphone)) directly or wirelessly coupled with the electronic device **101**.

[0057] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0058] The interface **177** may support one or more specified protocols to be used for the electronic

device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0059] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, an HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).

[0060] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0061] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0062] The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0063] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0064] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device **104** via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify or authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0065] The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may

support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0066] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

[0067] According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, an RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0068] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0069] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the external electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine

learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0070] FIG. **2** is a flowchart **200** for illustrating an operation of an electronic device for rearranging multiple items, based on a current time point according to an embodiment. FIG. **2** will be described in connection with FIG. **4A** to FIG. **4C**.

[0071] FIG. **4A** is a diagram for illustrating an operation of acquiring information included in a content displayed on a display of an electronic device, according to an embodiment.

[0072] FIG. **4B** is a diagram for illustrating an operation of arranging multiple items acquired based on information included in a content of an electronic device according to a priority and displaying the arranged multiple items, according to an embodiment.

[0073] FIG. **4C** is a diagram for illustrating an operation of rearranging, based on a changed priority, the arranged multiple items and displaying the rearranged multiple items, according to an embodiment.

[0074] Referring to FIG. **2**, in operation **210**, an electronic device (e.g., the electronic device **101** of FIG. **1** or the processor **120** of FIG. **1**) may acquire information included in a content displayed on a display (e.g., the display module **160** of FIG. **1**) based on the content satisfying a configured condition.

[0075] According to an embodiment, if the content includes at least one of time information, place information, or an event name, the electronic device may determine that the configured condition is satisfied, and may acquire the information included in the content. According to an embodiment, when the content is selected via a user input (e.g., a drag input or a long press input), the electronic device may acquire information included in the selected content.

[0076] According to an embodiment, as illustrated in FIG. **4A**, the electronic device may display a screen **410** including the content on the display. For example, the screen **410** may include an application execution screen.

[0077] According to an embodiment, if a content **411** included in the screen **410** satisfies the configured condition, the electronic device may acquire information included in the content **411**. For example, if the content **411** included in the screen **410** satisfies the configured condition, the electronic device may acquire information included in the content **411** even without receiving the user input.

[0078] According to an embodiment, the electronic device may acquire information included in the content **411** through the user input. For example, if the content **411** included in the screen **410** satisfies the configured condition or is selected by the user input, the electronic device may display a user interface **420** for storing of the content **411** in a partial area of the display so as to be overlaid with a part of the screen **410**.

[0079] According to an embodiment, when a user input to select the user interface **420** is received, the electronic device may highlight the acquired content **411**, or if the user selects the content **411**, the content **411** may be highlighted.

[0080] According to an embodiment, when the user input to select the user interface **420** is received again after highlighting the content **411**, the electronic device may acquire information included in the highlighted content **411**.

[0081] According to an embodiment, after highlighting the content **411**, the electronic device may display at least one pre-stored group, and upon reception of a user input to select a group (e.g., a schedule and/or a purchase list) in which information included in the content **411** is to be included, the electronic device may acquire the information included in the highlighted content **411** and may add the acquired information to the selected group. For example, the at least one pre-stored group may be generated by a user input, and each group may include information input by the user.

According to an embodiment, the at least one pre-stored group may include a group related to a

first schedule input by the user, a group related to a second schedule input by the user, and/or a group related to a purchase list input by the user. For example, the groups related to the schedules may include information input by the user via a calendar application, and the group related to the purchase list may include a goods list input by the user via a memo application.

[0082] According to an embodiment, the group related to the first schedule and/or the group related to the second schedule may include schedule information, place information, weather information, and/or an event name which are related to each schedule. For example, the group related to the purchase list may include information related to goods to be purchased, information on a place where the goods can be purchased, and/or discount information.

[0083] According to an embodiment, if there is one pre-stored group, the electronic device may highlight the content **411** and may display the pre-stored one group, and when a user input to select the displayed one group is received, the electronic device may add information included in the content **411** to the selected group.

[0084] According to an embodiment, an embodiment of acquiring information included in a displayed content will be described below with reference to FIG. 5A to FIG. 6.

[0085] According to an embodiment, the electronic device may acquire information included in a content by analyzing at least one of an image or text included in the content. According to an embodiment, the electronic device may group information included in contents acquired via multiple screens. For example, the electronic device may group information included in contents acquired via multiple screens into one schedule or one purchase list. For example, if a first content included in an execution screen of a first application displayed on the display satisfies the configured condition, the electronic device may acquire information included in the first content, and if a second content included in an execution screen of a second application displayed on the display concurrently with or after the first application satisfies the configured condition, the electronic device may acquire information included in the second content. According to an embodiment, if the first content or the second content includes at least one of time information, place information, and an event name, the electronic device may determine that the configured condition is satisfied.

[0086] According to an embodiment, in operation **220**, the electronic device may arrange, based on a first priority, a plurality of items acquired based on information included in the content so as to store the same in a memory (e.g., the memory **130** of FIG. 1). According to an embodiment, the electronic device may determine the first priority, based on a purpose of grouping the information. According to an embodiment, the grouping purpose may be related to a type of a service to be provided to the user via grouping of the information. For example, the electronic device may identify whether the service to be provided is related to an itinerary, a meeting schedule, and/or a purchase list, based on the text included in the information input by the user and/or the text and/or image included in the information acquired via the displayed content.

[0087] According to an embodiment, the first priority may be preconfigured for each grouping purpose. For example, if information is grouped in relation to an itinerary, the first priority may be according to the itinerary. According to an embodiment, if information is grouped in relation to a meeting schedule, the first priority may be configured according to importance of a meeting progress. According to an embodiment, if information is grouped in relation to the purchase list, the first priority may be configured according to importance for purchasing of goods, such as a purchase place for goods to be purchased or a list of goods.

[0088] According to an embodiment, the priority may be changed according to a time flow and/or occurrence of a new event (e.g., a change in importance, an information change, and a goods purchase).

[0089] According to an embodiment, contents related to the priority will be described in more detail in [Table 1] and operation **230** below.

[0090] According to an embodiment, the electronic device may perform classification into the

plurality of items, based on details of information included in the content. For example, the electronic device may acquire the plurality of items, based on information included in the first content acquired via the execution screen of the first application and information included in the second content acquired via the execution screen of the second application.

[0091] According to an embodiment, the electronic device may group information included in the contents acquired via analysis of at least one of text or images included in the contents into one schedule or one purchase list, and may classify the grouped information into the plurality of items.

[0092] According to an embodiment, when acquiring information included in the first content via the first application screen and acquiring information included in the second content via the second application screen, the electronic device may group the information included in the first content and the information included in the second content into one schedule or one purchase list. For example, the electronic device may perform grouping into one schedule or one purchase list, based on keywords and/or schedule information of the information included in the first content and the information included in the second content.

[0093] According to an embodiment, grouping of information refers to classification of acquired information, and the electronic device may group information, based on relevance between respective pieces of information. For example, if at least some of the plurality of pieces of the information included in each content is the same, or similar in time and/or location even if the information is not the same, the electronic device may determine that there is relevance between respective pieces of information.

[0094] According to an embodiment, if information included in a content included in a screen of an airline application and location information and/or schedule information of information included in a content included in a screen of a social network service (SNS) application are correlated with each other, the electronic device may group the information into one schedule.

[0095] According to one embodiment, information of goods, which is displayed multiple times on the screen of the SNS application or is included in a content included in a screen displayed by the user's search, and a content, which is displayed multiple times on a screen of a shopping application or is included in the screen displayed by the user's search, indicate that the user has a purchase intention, and the electronic device may group the information and the content into the purchase list. According to an embodiment, when the content displayed on the screen of the SNS application includes information related to goods for a travel to a specific place, and a content displayed on an Internet application screen includes information related to the goods for the travel to the specific place, the electronic device may group information included in the two contents into one purchase list, based on relevance of the place.

[0096] According to an embodiment, the plurality of items may include schedules (e.g., a flight date/time and a meeting date/time), places (e.g., accommodation, tourist attractions, meeting places, parking lots, and restaurants), to-do lists, attendees, weather, discount information, and/or checklists.

[0097] According to an embodiment, the electronic device may classify the grouped information into the plurality of items according to a grouping purpose. For example, if grouped into the itinerary, the electronic device may classify the grouped information into the plurality of items including a flight date/time, accommodation, a tourist destination, weather, a restaurant, a traffic condition, and/or a checklist (e.g., preparations). According to an embodiment, if grouped into the meeting schedule, the electronic device may have grouped information including a meeting date/time, a meeting place, attendees, a traffic condition, and/or a checklist (e.g., tasks and materials). According to an embodiment, if grouped into the purchase list, the electronic device may classify grouped information into the plurality of items including a purchase place and/or a purchase list. Types of the plurality of items described above are merely examples and are not limited thereto.

[0098] According to an embodiment, the electronic device may acquire the plurality of items by

further including information received from an external server (e.g., the server **108** of FIG. **1**) via a communication module (e.g., the communication module **190** of FIG. **1**). For example, the electronic device may receive weather information, time information, flight information, place information (e.g., parking lot information and restaurant information), and/or discount information from the external server via the communication module, and may acquire the plurality of items by further including the received information. For example, the electronic device may perform grouping into one schedule or one purchase list by further including information received from the external server, and may classify the grouped information into the plurality of items. Accordingly, the plurality of items may be acquired by further including information that is not displayed on the display.

[0099] According to an embodiment, the electronic device may receive a user input to input information via the display, and may acquire the plurality of items by further including information input via the user input. For example, when the electronic device receives a user input to input a schedule via the calendar application displayed on the display or for inputting a purchase list through the memo application, the electronic device may perform grouping into one schedule or one purchase list by further including information input via the user input and may classify the grouped information into the plurality of items.

[0100] According to an embodiment, the information acquired via the content displayed on the display, the information received from the external server, and/or the information acquired by the user input may be classified into the plurality of items (e.g., related items in Table 1) as shown in [Table 1] below according to the type of information.

TABLE-US-00001 TABLE 1 Associated linkage Extracted information Usage schemes fields
Detailed places and locations Specifying detailed locations Place Traffic situations Recommending an alternative place in consideration of a traffic situation Parking situations Recommending a nearby parking lot due to full parking lots Phone number/business Using as detailed place hours information Weather Changing a place Changing a schedule Applying a change in a Schedule booked schedule Nearby must-visit Specifying an itinerary in restaurant locations, more detail, such as applying recommended menu a detailed schedule within the items, business hours, and itinerary and adding a nearby event information detailed schedule Recommending a menu item Recommending an action to Detailed Recommending payment be taken during a schedule schedules discount Related checklists Specifying and To do Related product discount recommending (preparation) information a preparation item Detailed Related product discount Changing a to-do priority actions information according to a discount period Changes via an Recommendations for email message recognizing key changes Changes of attachment before attending a schedule Unread related emails event

[0101] According to an embodiment, referring to [Table 1], the electronic device may use information in different ways according to a type of acquired information.

[0102] According to an embodiment, the electronic device may arrange the plurality of items, based on the first priority. For example, the electronic device may arrange the plurality of items, based on the preconfigured first priority. According to an embodiment, in a case of the plurality of travel-related items, the first priority may be based on the itinerary. For example, the itinerary may be arranged so that an item related to a departure schedule of the travel has the highest priority, and items related to schedules to be progressed for the travel have priorities in chronological order.

[0103] According to an embodiment, in a case of the plurality of items related to the meeting, the first priority may be configured according to importance, and the electronic device may arrange a date/time item related to the meeting to have the highest priority, a place item to have a second highest priority, and an attendee item to have a subsequent priority.

[0104] According to an embodiment, in a case of the plurality of items related to the purchase list, the electronic device may arrange the first priority so that a place item, at which goods included in the purchase list can be purchased, has the highest priority and a list of goods to be purchased has a

subsequent priority.

[0105] According to an embodiment, when the user input to check the itinerary or the purchase list is received, the electronic device may display the plurality of items arranged based on the first priority, as illustrated in FIG. 4B.

[0106] Referring to FIG. 4B, the electronic device may display a screen **430** for checking the itinerary. According to an embodiment, the screen **430** may include the plurality of items **431** and **432** arranged based on the first priority. According to an embodiment, the plurality of items **431** and **432** may include links for providing summary information related to the items and/or detailed information related to the items, respectively. For example, the links may include icons **41**, **42**, **43**, and **44** of applications that provide the detailed information related to the items.

[0107] According to an embodiment, the first priority may be based on a sequence of an itinerary according to time, and the departure flight date/time item **431** related to a departure schedule may be arranged to have the highest priority, and the subsequent priority item **432** may include an accommodation item and a tourist destination item which are sequentially arranged.

[0108] According to an embodiment, in operation **230**, if a current time point satisfies the configured condition, the electronic device may rearrange the plurality of items, based on the second priority. According to an embodiment, the electronic device may rearrange the plurality of items over time (e.g., a time line). For example, the configured condition may be arrival of a specific time point related to the itinerary. According to an embodiment, monitoring is performed by comparing a current time, which is measured by the electronic device itself or received from a time-related server, with time point information related to the itinerary stored for each item, and if arrival of the time point information related to the itinerary is determined, it may be determined that the configured condition is satisfied.

[0109] According to an embodiment, the second priority may be based on a matter related to the itinerary, which is required to be checked at a current time point, and may correspond to the plurality of items rearranged so that an item related to the matter required to be checked have the highest priority. For example, if there is a matter required to be checked at a specific time point (e.g., 3 days before a schedule, a scheduled day, and 3 hours before the schedule), which is related to the itinerary, for each item, the electronic device may, when the specific time point arrives, rearrange the plurality of items and provide a notification, based on the second priority in which a priority of a corresponding item has the highest priority. According to an embodiment, as for orders after the highest priority in the second priority, items related to the matter required to be checked may be sequentially arranged over time or the pre-arranged sequence based on the first priority may be maintained.

[0110] According to an embodiment, the electronic device may acquire information required to be checked at the specific time point related to the itinerary, via information input by the user or information acquired from the content. For example, the electronic device may acquire text information stating that, in a case of cancellation via a content included in a screen of a booking application, a time point at which 100% refund is available is one week before a booked schedule, and information on the time point may be stored in the memory. According to an embodiment, the electronic device may perform monitoring of comparing a current time point with time point information of one week before a booked accommodation schedule, and may rearrange the plurality of items so that items related to accommodation have the highest priority one week before the booked accommodation schedule.

[0111] According to an embodiment, as for the second priority, an item related to the matter required to be checked may have the highest priority, and for subsequent items, items related to matters required to be checked over time may be sequentially arranged, or the pre-arranged sequence based on the first priority may be maintained except for the item related to the matter required to be checked.

[0112] According to an embodiment, if the current time point is before a first period (e.g., 3 days

before the start of the schedule) of a time point related to one schedule, the electronic device may rearrange the plurality of items, based on the second priority, and if the current time point is before a second period (e.g., 1 day before the start of the schedule) that is shorter than the first period of the time point related to one schedule, the electronic device may rearrange the plurality of items, based on a third priority. According to an embodiment, rearranging of the plurality of items related to a travel or a meeting over time will be described below with reference to FIG. 8A to FIG. 10D.

[0113] In the above, it has been described that operation **230** is rearranging of the plurality of items which have been arranged based on the first priority over time via operation **220**. However, according to an embodiment, the plurality of items rearranged over time may be rearranged as more time passes, and hereinafter, the plurality of items rearranged by an information change in operation **330** of FIG. 3 may be rearranged over time.

[0114] In the above, it has been described that, after operation **210** of acquiring information, operation **220** of arranging the plurality of items and operation **230** of rearranging the plurality of arranged items are performed. However, according to an embodiment, acquiring of information may be performed even after the plurality of items are arranged based on the first priority, and acquiring of information may be performed also after the plurality of arranged items are arranged based on the second priority.

[0115] According to an embodiment, in operation **240**, the electronic device may display the plurality of rearranged items on the display.

[0116] According to an embodiment, when the plurality of items are rearranged, the electronic device may control the display to display a notification of the rearrangement, and when a user input to select the notification is received, the electronic device may control the display to display the plurality of rearranged items. According to an embodiment, displaying of the notification of the rearrangement of the plurality of items and displaying of the rearranged items according to a notification selection will be described below with reference to FIG. 7A to FIG. 7D.

[0117] For example, as illustrated in FIG. 4C, when the user input to check information changed in relation to the itinerary is received, the electronic device may display a screen **440** for checking the itinerary. According to an embodiment, the screen **440** may include the plurality of items **441** and **442** arranged based on the second priority.

[0118] According to an embodiment, the second priority may be based on the matter related to the itinerary, which is required to be checked at the current time point, and may correspond to the plurality of items **441** and **442** rearranged so that the traffic item **441** related to the matter required to be checked has the highest priority. According to an embodiment, as for the second priority, the item **441** related to the matter required to be checked may have the highest priority, and for the subsequent item **442**, items related to matters required to be checked over time may be sequentially arranged, or the pre-arranged sequence based on the first priority may be maintained.

[0119] FIG. 3 is a flowchart **300** for illustrating an operation of an electronic device for rearranging the plurality of items, based on a change of information according to an embodiment.

[0120] Referring to FIG. 3, in operation **310**, an electronic device (e.g., the electronic device **101** of FIG. 1 or the processor **120** of FIG. 1) may acquire information included in a content displayed on a display (e.g., the display module **160** of FIG. 1) if the content satisfies a configured condition.

[0121] Since operation **310** of FIG. 3 is the same as operation **210** of FIG. 2 described above, a redundant description will be omitted.

[0122] According to an embodiment, in operation **320**, the electronic device may arrange, based on a first priority, the plurality of items acquired based on information included in the content so as to store the same in a memory (e.g., the memory **130** of FIG. 1).

[0123] Operation **320** of FIG. 3 is the same as operation **220** of FIG. 2 described above, and will be thus omitted.

[0124] According to an embodiment, in operation **330**, when information related to at least one of the plurality of items is changed, the electronic device may update information included in the at

least one item, and may rearrange the plurality of updated items based on a second priority.

[0125] According to an embodiment, when an event occurs in which the information included in the at least one of the plurality of items is changed, the electronic device may rearrange the plurality of items so that the item in which the information is changed has the highest priority.

[0126] According to an embodiment, when the electronic device receives the changed information for at least one of the plurality of items from an external server (e.g., the server **108** of FIG. **1**) via a communication module (e.g., the communication module **190** of FIG. **1**), the electronic device may identify that the event has occurred and may rearrange the plurality of items so that the item for which the event has occurred has the highest priority. For example, when change information, such as change information of a weather item, change information of a place item (e.g., parking lot information), or discount information, is received from the external server, the electronic device may change the corresponding item to have the highest priority.

[0127] According to an embodiment, in a case of the plurality of items related a purchase list, when discount information is received from the external server in a state where the place item has the highest priority and a purchase goods list item has a subsequent priority according to the first priority, the electronic device may change information of a corresponding item while maintaining the sequence of the item. For example, the electronic device may change information on the place item having the highest priority to information on a place where a discount is performed, and may change priority of goods in the purchase goods list item so that discounted goods has a higher priority.

[0128] According to an embodiment, when information included in the content displayed via the display is changed, the electronic device may identify that an event has occurred, and may rearrange the plurality of items so that an item for which the event has occurred has the highest priority. For example, in a case of the plurality of items related to a schedule, if the electronic device acquires information included in the content displayed on the display, and the information included in the content is for changing of at least one piece of information in the plurality of pre-stored items, the electronic device may rearrange the plurality of items so that an item, in which information has been changed, has the highest priority. For example, if time information of the schedule (e.g., exhibition) on an application (e.g., Internet) execution screen is different from pre-stored time item information, the electronic device may rearrange the plurality of items so that the time item has the highest priority. According to an embodiment, the electronic device may change the pre-stored time item information to time information included in the content, or may determine whether to change time item information according to a user input to select the pre-stored time item information or the time information included in the content. According to an embodiment, in a case of the plurality of items related to the purchase list, when at least one of the plurality of goods included in the purchase goods list is purchased online, the electronic device may acquire information related to the purchased goods from a payment screen displayed on the display, may delete the purchased goods from among the plurality of goods included in the purchase goods list, and may change a sequence of goods in the purchase goods list.

[0129] According to an embodiment, when a user input to change at least one piece of information among the pre-stored the plurality of items is received, the electronic device may identify that an event has occurred and may rearrange the plurality of items so that an item in which the event has occurred has the highest priority. For example, in the case of the plurality of items related to the schedule, when a user input to change time information or place information is received, the electronic device may rearrange the plurality of items so that an item in which information has been changed has the highest priority. According to an embodiment, in the case of the plurality of items related to the purchase list, when a user input to delete some goods from the purchase goods list is received, the electronic device may delete some goods from the purchase goods list and change the sequence of the goods.

[0130] According to an embodiment, based on change information acquired via the content

displayed on the display, change information received from the external server, and/or change information by the user input, items and/or item information may be changed as shown in [Table 2] below.

TABLE-US-00002 TABLE 2 Recommended Detailed types Associated information
recommendations Change Weather .fwdarw. place Changing a place due to a Changing weather
change information Parking information .fwdarw. Recommending a nearby of a place parking lot
due to full parking basic item lots Information of Changing a schedule due to a surroundings
.fwdarw. schedule traffic situation/a change in business hours Changing schedule Applying a
schedule change information .fwdarw. schedule via a message and an e-mail for a change of a
schedule Specifying Recommended menu Presenting recommendations Specifying information
.fwdarw. detailed useful when carrying out a information of actions schedule, via recommended
existing items menu items of must-visit restaurants Recommending must-visit Recommending
destinations restaurants .fwdarw. or specific places/courses to detailed actions visit Payment
discount Specifying actions via information .fwdarw. payment discount information detailed
actions when carrying out a schedule Prioritization Discount information Changing purchase
priorities Changing in a to-do list based on priority of an discount product information existing
item Relevant up-to-date Changing a priority order of information that has not unchecked
important been checked (emails and information among documents) information to be checked
before starting a schedule

[0131] In the above, it has been described that operation **330** is rearranging of the plurality of items arranged based on the first priority via operation **320**, according to a change of information. However, according to an embodiment, when at least one piece of information from among the plurality of items rearranged in operation **230** of FIG. 2 is changed, arrangement may be performed according to operation **330**, and after the plurality of items are rearranged according to the change of the information, when a new information change occurs, rearrangement may be performed based on the new information change.

[0132] According to an embodiment, in operation **340**, the electronic device may display the plurality of rearranged items on the display.

[0133] Operation **340** of FIG. 3 is the same as operation **240** of FIG. 2 described above, and will be thus omitted.

[0134] According to an embodiment, rearrangement of the plurality of items over time, which is operation **230** of FIG. 2, and rearrangement of the plurality of items due to the information change, which is operation **330** of FIG. 3, may be performed twice or more with respect to one schedule or one purchase list, and this will be described with reference to FIG. 8A to FIG. 11C.

[0135] FIG. 5A is a diagram for illustrating an operation of acquiring information included in a content displayed on a display of an electronic device, according to an embodiment.

[0136] Referring to FIG. 5A, an electronic device (e.g., the electronic device **101** of FIG. 1 or the processor **120** of FIG. 1) may display an execution screen of an application (e.g., an SNS application or an Internet application) on a display (e.g., the display module **160** of FIG. 1).

[0137] According to an embodiment, the electronic device may acquire information included in a content (e.g., an image or text) included in the execution screen of the application. For example, the electronic device may acquire restaurant-related information **510** including information on surroundings, business hours, location information, phone number, and/or a combination of recommended menu items included in the displayed content.

[0138] According to an embodiment, the electronic device may recognize a positive content or a negative content via analysis of the content (e.g., an image or text) and may classify information included in the content. For example, if the content includes text, such as “recommended”, “must-visit restaurant”, and/or “satisfied”, the electronic device may recognize the same as a positive content so as to acquire information, and if the content includes text or an image, such as star symbols (e.g., .star-solid..star-solid..star-solid.★ ★), the electronic device may acquire information

by selecting a positive content, based on the number of stars in the star symbols.

[0139] FIG. 5B is a diagram for illustrating an operation of acquiring information included in a content displayed on the display of the electronic device, according to an embodiment.

[0140] Referring to FIG. 5B, the electronic device may display an execution screen of an application (e.g., an SNS application or an Internet application) on the display.

[0141] According to an embodiment, the electronic device may acquire information included in a content (e.g., an image or text) included in the execution screen of the application. For example, the electronic device may acquire exhibition-related information **520** including an event name, schedule information including date and time, and/or location information, which are included in the displayed content.

[0142] FIG. 5C is a diagram for illustrating an operation of acquiring information included in a content displayed on the display of the electronic device, according to an embodiment.

[0143] Referring to FIG. 5C, the electronic device may display an execution screen of an application (e.g., an SNS application or an Internet application) on the display.

[0144] According to an embodiment, the electronic device may acquire information included in a content (e.g., an image or text) included in the execution screen of the application. For example, the electronic device may acquire discount information **530** including a discount period, a discount store location, a discount product, and/or a discount rate, which are included in the displayed content.

[0145] FIG. 5D is a diagram for illustrating an operation of acquiring information included in a content displayed on the display of the electronic device, according to an embodiment.

[0146] Referring to FIG. 5D, the electronic device may display an execution screen of an application (e.g., a weather application or an Internet application) on the display.

[0147] According to an embodiment, the electronic device may acquire information included in a content (e.g., an image or text) included in the execution screen of the application. For example, the electronic device may acquire weather information **540** including location information, time information, weather information, temperature information, wind direction information, wind speed information, precipitation information, and/or humidity information, which are included in the displayed content. For example, the acquired weather information **540** may be related to one schedule.

[0148] FIG. 5E is a diagram for illustrating an operation of acquiring information included in a content displayed on the display of the electronic device, according to an embodiment.

[0149] Referring to FIG. 5E, the electronic device may display an execution screen of an application (e.g., an Internet application, a document application, or a memo application) on the display.

[0150] According to an embodiment, the electronic device may acquire information included in a content (e.g., an image or text) included in the execution screen of the application. For example, the electronic device may acquire a checklist **550** including a list of preparations included in the displayed content. For example, the acquired checklist **550** may be related to one schedule or one purchase list.

[0151] As described above, the electronic device may acquire information that is highly related to a user's interest by acquiring information included in the content displayed on the display.

[0152] FIG. 6 is a diagram for illustrating an operation of acquiring information included in a content displayed on a display of an electronic device, according to an embodiment.

[0153] Referring to FIG. 6, an electronic device (e.g., the electronic device **101** of FIG. 1 or the processor **120** of FIG. 1) may display an application execution screen **610** on a display (e.g., the display module **160** of FIG. 1). According to an embodiment, if a content that satisfies a configured condition is included in the application execution screen **610**, the electronic device may display a user interface **620** for acquiring information included in the content, in a partial area of the display so as to be overlaid with a part of the application execution screen **610**.

[0154] According to an embodiment, when a user input **621** for selecting the user interface **620** is received, the electronic device may highlight a content **611** that satisfies the configured condition, and may display a message **630** (e.g., “Do you want to save relevant information?”) for asking whether to acquire information included in the highlighted content **611** around the user interface **620**.

[0155] According to an embodiment, after displaying of the message **630**, when a user input to select the user interface **620** is received, the electronic device may acquire information included in the highlighted content **611**.

[0156] In the above, it has been illustrated and described that the user interface **620** is displayed, and then the content **611** satisfying the configured condition is highlighted. However, according to an embodiment, in a case where a user selects a content for which information is to be acquired, when a user input (e.g., a drag or a long press) for selecting the content is received, the electronic device may highlight the selected content and display the user interface **620** for acquiring information included in the content. According to an embodiment, when the user input to select the content is received, a message for asking whether to acquire information may not be displayed.

[0157] According to an embodiment, when multiple groups (e.g., an information group related to a travel to Spain, or an information group related to a meeting) in which information has been grouped are stored in a memory (e.g., the memory **130** of FIG. 1), the electronic device may display a list of the stored multiple groups in addition to or by replacing a message **630** display operation for asking whether to acquire information, and if a user input to select one group among the multiple groups included in the list is received, the electronic device may add information included in the highlighted content to the selected group. According to an embodiment, when the user input to select the content is received, and then the user input to select the user interface **620** is received, the electronic device may display the list of multiple groups around the user interface **620**, and when the user input to select one group from among the multiple groups included in the list is received, the electronic device may add information included in the highlighted content to the selected group.

[0158] FIG. 7A is a diagram for illustrating an operation of an electronic device when the plurality of items are rearranged based on a changed priority, according to an embodiment.

[0159] Referring to FIG. 7A, if priorities of the plurality of items arranged based on a first priority is changed so as to be arranged based on a second priority, the electronic device (e.g., the electronic device **101** of FIG. 1 or the processor **120** of FIG. 1) may display, on a part of the display, a notification **710** (e.g., “A change has occurred in the input schedule.” or “There is a new event related to the input schedule.”) indicating that the priorities have been changed. For example, the electronic device may display a notification **710** in a pop-up form. For example, the priorities may be changed over time or based on at least a partial change of information included in the plurality of items.

[0160] According to an embodiment, after the notification **710** is displayed, when a user input **711** for checking changed information is received, an application execution screen for checking of the changed information may be displayed as illustrated in FIG. 7B.

[0161] FIG. 7B is a diagram for illustrating an operation of the electronic device when the plurality of items are rearranged based on changed priorities, according to an embodiment.

[0162] Referring to FIG. 7B, the electronic device may display an execution screen **720** of an application (e.g., a calendar application), for checking of the changed information.

[0163] For example, if priorities of the plurality of items related to one schedule is changed, a notification **722** may be displayed on a schedule **721** in which the priorities have been changed.

[0164] According to an embodiment, when the notification **722** is displayed on the schedule **721** in which the priorities have been changed, and then the user input to check the changed information is received, the application execution screen to check for the changed information may be displayed as illustrated in FIG. 7C.

[0165] FIG. 7C is a diagram for illustrating an operation of the electronic device when the plurality of items are rearranged based on changed priorities, according to an embodiment.

[0166] Referring to FIG. 7C, the electronic device may display the plurality of items **730** related to the selected schedule, based on the priorities. For example, the electronic device may change priorities of the plurality of items over time and may display the plurality of items **730** rearranged based on the changed priorities.

[0167] According to an embodiment, the electronic device may change the priorities over time so that an item **731** required to be checked at a current time has the highest priority. For example, if the item required to be checked on a corresponding day is the restaurant item **731**, the electronic device may perform rearrangement so that the restaurant item **731** has the highest priority, and a weather item and a checklist item, which had to be checked 3 days ago, have subsequent priorities.

[0168] FIG. 7D is a diagram for illustrating an operation of the electronic device when the plurality of items are rearranged based on changed priorities, according to an embodiment.

[0169] Referring to FIG. 7D, the electronic device may display the plurality of items **740** related to the selected schedule, based on the changed priorities. For example, the electronic device may change the priorities of the plurality of items as information is changed and may display the plurality of items rearranged based on the changed priorities.

[0170] According to an embodiment, as information of at least one item among the plurality of items is changed, the electronic device may change the priorities so that the item, the information of which has been changed, has the highest priority. For example, when weather information is changed, the electronic device may rearrange a weather item **741** to have the highest priority. According to an embodiment, the electronic device may maintain the previously arranged the plurality of items (e.g., the plurality of items **730** of FIG. 7C) for the subsequent priorities.

[0171] FIG. 8A is a diagram for illustrating electronic device operations of acquiring information, arranging the plurality of items based on the acquired information according to a priority, and displaying the plurality of arranged items, according to an embodiment.

[0172] Referring to FIG. 8A, an electronic device (e.g., the electronic device **101** of FIG. 1 or the processor **120** of FIG. 1) may acquire and group **810** information related to an itinerary. According to an embodiment, when a content displayed on a display satisfies a configured condition, the electronic device may acquire and group information included in the content. For example, when a flight ticket is booked via a first application (e.g., an airline application or a booking agency application), the electronic device may acquire destination information and/or date/time information included in an execution screen of the first application.

[0173] According to an embodiment, when booking accommodation via a second application (e.g., an accommodation application or the booking agency application), and searching for a tourist destination by using a third application (e.g., an SNS application, an Internet application, the booking agency application, or a map application) and a fourth application (e.g., the SNS application, the Internet application, the booking agency application, and the map application), the electronic device may acquire place information and/or date/time information included in execution screens of the second application to the fourth application.

[0174] According to an embodiment, the electronic device may group the acquired information into one group depending on a purpose. For example, the electronic device may group the acquired information into one schedule. According to an embodiment, the electronic device may perform grouping by further adding, to the information acquired via the content, information received from an external server and information acquired by a user input.

[0175] According to an embodiment, the electronic device may acquire the tourist destination and the travel date/time via information acquired from the execution screen of the first application, and may acquire movement information of a user according to the user's field of interest and/or date via the execution screens of the second application to the fourth application, and may group the acquired information into one itinerary. For example, the electronic device may acquire, via the

acquired information, information that the tourist destination is Spain and the itinerary is from November 1st to November 14.

[0176] According to an embodiment, the electronic device may classify the grouped information into the plurality of items. For example, the plurality of items may include a schedule (e.g., a flight date/time and a meeting date/time) item, a place (e.g., accommodations, tourist attractions, meeting places, parking lots, and restaurants) item, a to-do list item, an attendee item, a weather item, a discount information item, and/or a checklist (e.g., preparations) item.

[0177] According to an embodiment, the electronic device may arrange **820** information, based on priorities. For example, the electronic device may arrange the plurality of items obtained by classifying the grouped information, based on a configured first priority.

[0178] According to an embodiment, in a case of the plurality of travel-related items, the first priority may be based on the itinerary. For example, the itinerary may be arranged so that an item related to a departure schedule of the travel has the highest priority, and items related to schedules to be progressed for the travel have priorities in chronological order.

[0179] According to an embodiment, when the electronic device is able to perform grouping into one group (e.g., a schedule or a purchase list) via information acquired for a certain period (e.g., before September 1), the electronic device may classify the acquired information into the plurality of items and arrange the same according to the first priority, and may provide the plurality of arranged items **830** to the user. According to an embodiment, each of the plurality of items **830** may include a link for providing of summary information related to the item and/or detailed information related to the item. For example, the link may include an icon of an application that provides detailed information related to the item.

[0180] According to an embodiment, the electronic device may rearrange priorities over time or according to a change of information, which will be described below with reference to FIG. **8B** to FIG. **8G**.

[0181] FIG. **8B** is a diagram for illustrating electronic device operations of rearranging, based on priorities, the plurality of items and displaying the plurality of rearranged items, according to an embodiment. For example, FIG. **8B** illustrates a case in which the priorities of the plurality of items are changed according to an information change.

[0182] Referring to FIG. **8B**, when weather information (e.g., sunny->rain) changed in relation to the schedule (e.g., November 1 to November 14) is received from an external server before the schedule (e.g., October 1), the electronic device may change a method related to the tourist destination to an indoor tourist destination method (e.g., a city bus tour), may rearrange the plurality of items so that a changed tourist destination item **840** has the highest priority, and may indicate that the change of the tourist destination method is recommended due to the change in weather.

[0183] FIG. **8C** is a diagram for illustrating electronic device operations of rearranging, based on priorities, the plurality of items and displaying the plurality of rearranged items, according to an embodiment. For example, FIG. **8C** illustrates a case in which the priorities of the plurality of items are changed over time.

[0184] Referring to FIG. **8C**, when a deadline for changing a booked accommodation date arrives, the electronic device may rearrange the plurality of items so that an accommodation item **850** has the highest priority, and may display information (e.g., a scheduled day or two days later) related to the deadline for changing the booked date. For example, the electronic device may identify that the deadline for changing the booked accommodation date has arrived, based on booked date change deadline information acquired from a content displayed during accommodation booking and booked date change deadline information received from the external server related to the accommodation application.

[0185] FIG. **8D** is a diagram for illustrating electronic device operations of rearranging, based on priorities, the plurality of items and displaying the rearranged the plurality of items, according to an

embodiment. For example, FIG. **8B** illustrates a case in which the priorities of the plurality of items are changed according to an information change.

[0186] Referring to FIG. **8D**, when information related to a changed flight departure time is received from the external server, the electronic device may rearrange the plurality of items so that a departure item **860** has the highest priority, and may display the changed information (e.g., a delay) of the flight departure time.

[0187] FIG. **8E** is a diagram for illustrating electronic device operations of rearranging, based on priorities, the plurality of items and displaying the plurality of rearranged items, according to an embodiment. For example, FIG. **8E** illustrates a case in which the priorities of the plurality of items are changed over time.

[0188] Referring to FIG. **8E**, when a deadline for discounting an admission ticket for a tourist destination arrives, the electronic device may rearrange the plurality of items so that a tourist destination item **870** has the highest priority, and may display discount information (e.g., a period and/or a discount rate). For example, the electronic device may identify that the discount available deadline has arrived, based on booked date change deadline information acquired from a content displayed during a tourist destination search and booked date change deadline information received from the external server.

[0189] FIG. **8F** is a diagram for illustrating electronic device operations of rearranging, based on priorities, the plurality of items and displaying the plurality of rearranged items, according to an embodiment. For example, FIG. **8F** illustrates a case in which the priorities of the plurality of items are changed over time.

[0190] Referring to FIG. **8F**, the electronic device may rearrange the plurality of items so that an item including information to be checked by the user has the highest priority, based on a scheduled itinerary at the current time point (e.g., November 6) during the travel. For example, when the user desires to check a place to take a bus to visit the tourist destination according to the scheduled itinerary, the electronic device may rearrange the plurality of items so that a tourist destination item **880** has the highest priority, and may display that the place to take the bus needs to be checked.

[0191] According to an embodiment, the electronic device may change the content of the tourist destination item so as to enable viewing of a record, such as a photo taken in relation to the tourist destination visited according to the scheduled itinerary.

[0192] FIG. **8G** is a diagram for illustrating electronic device operations of rearranging, based on priorities, the plurality of items and displaying the plurality of multiple items, according to an embodiment. For example, FIG. **8G** illustrates a case in which the priorities of the plurality of items are changed over time.

[0193] Referring to FIG. **8G**, the electronic device may rearrange the plurality of items so that an item, which includes information to be checked by the user one day (e.g., November 13) before the last day (e.g., November 14) of the travel, has the highest priority. For example, if the user desires to check a method to get to an airport, the electronic device may rearrange the plurality of items so that a flight item **890** has the highest priority, and may display that the method to get to the airport needs to be checked.

[0194] According to an embodiment, the electronic device may change the content of the tourist destination item so as to enable viewing of a record, such as a photo taken in relation to the tourist destination visited according to the scheduled itinerary.

[0195] According to an embodiment, when the itinerary ends (e.g., on November 16), the electronic device may terminate **895** updating of information on the itinerary.

[0196] The aforementioned changing of the priorities, illustrated in FIG. **8A** to FIG. **8G**, is merely an example, and a reason, number of times, and/or sequence of the priority change are not limited thereto.

[0197] FIG. **9A** is a diagram for illustrating electronic device operations of acquiring information, arranging the plurality of items based on the acquired information according to a priority, and

displaying the plurality of arranged items, according to an embodiment.

[0198] Referring to FIG. 9A, an electronic device (e.g., the electronic device **101** of FIG. 1 or the processor **120** of FIG. 1) may acquire and group **910** information related to a meeting schedule. According to an embodiment, when a content displayed on a display satisfies a configured condition, the electronic device may acquire and group information included in the content. For example, a first application (e.g., an email application and an Internet application) may receive a user input to store an attachment included in a meeting-related content.

[0199] According to an embodiment, the electronic device may acquire (e.g., extract) the meeting-related content via an execution screen of the first application. For example, the electronic device may acquire meeting place information, meeting time information, and/or meeting subject information which are included in the body of an email.

[0200] According to an embodiment, when a user input to check the progress of a task (to-do) (e.g., preparing meeting materials) in a second application (e.g., a collaboration application) is received, the electronic device may analyze a content (e.g., text or an image) included in an execution screen of a third application so as to check the progress of the task.

[0201] According to an embodiment, the electronic device may group the acquired information into one group depending on a purpose. For example, the electronic device may group the acquired information into one schedule. According to an embodiment, the electronic device may perform grouping by further adding, to the information acquired via the content, information received from an external server and information acquired by a user input.

[0202] According to an embodiment, the electronic device may classify the grouped information into the plurality of items. For example, the plurality of items may include a schedule (e.g., a meeting date/time) item, a to-do list item, a latest meeting minutes item, and/or an attendee item.

[0203] According to an embodiment, the electronic device may arrange **920** information, based on priorities. For example, the electronic device may arrange the plurality of items obtained by classifying the grouped information, based on a configured first priority.

[0204] According to an embodiment, in a case of the plurality of meeting-related items, the first priority may be according to importance required for the progress of a meeting. For example, arrangement may be performed so that the schedule item, which is information required to attend a meeting, has the highest priority, and the to-do item and the latest meeting minutes item, which are information required for proceeding with the meeting, have subsequent priorities in sequence.

[0205] According to an embodiment, when the electronic device is able to perform grouping into one group (e.g., a schedule or a purchase list) via information acquired for a certain period (e.g., before September 12), the electronic device may classify the acquired information into the plurality of items and arrange the same according to the first priority, and may provide the plurality of arranged items **930** to a user. According to an embodiment, each of the plurality of items **930** may include a link for providing of summary information related to the item and/or detailed information related to the item. For example, the link may include an icon of an application that provides detailed information related to the item.

[0206] According to an embodiment, the electronic device may rearrange priorities over time or according to a change of information, which will be described below with reference to FIG. 9B to FIG. 9D.

[0207] FIG. 9B is a diagram for illustrating electronic device operations of rearranging, based on priorities, the plurality of items and displaying the plurality of rearranged items, according to an embodiment. For example, FIG. 9B illustrates a case in which the priorities of the plurality of items are changed according to an information change.

[0208] Referring to FIG. 9B, when information related to a changed meeting time is received from the external server, the electronic device may rearrange the plurality of items so that a schedule item **940** has the highest priority, and may display changed meeting schedule information and/or a reason for the change.

[0209] FIG. **9C** is a diagram for illustrating electronic device operations of rearranging, based on priorities, the plurality of items and displaying the plurality of rearranged items, according to an embodiment. For example, FIG. **9C** illustrates a case in which the priorities of the plurality of items are changed over time.

[0210] Referring to FIG. **9C**, when a specific time arrives before an execution deadline of an unperformed task among tasks required to be performed in the to-do list included in a to-do item **951**, the electronic device may rearrange the plurality of items so that a to-do item **950** has the highest priority, and may display the remaining deadline.

[0211] According to an embodiment, in the to-do item **951** including the to-do list, the electronic device may display a performed task as being deleted from the list, so as to indicate that the task has been completed.

[0212] FIG. **9D** is a diagram for illustrating electronic device operations of rearranging, based on priorities, the plurality of items and displaying the plurality of rearranged items, according to an embodiment. For example, FIG. **9D** illustrates a case in which the priorities of the plurality of items are changed according to an information change.

[0213] Referring to FIG. **9D**, when a new material (e.g., e-mail) related to the meeting is acquired from the external server, the electronic device may rearrange the plurality of items so that a material item **960** has the highest priority, and may display a title of the newly acquired material and/or a sender of the material.

[0214] FIG. **10A** is a diagram for illustrating electronic device operations of acquiring information, arranging the plurality of items based on the acquired information according to a priority, and displaying the plurality of arranged items, according to an embodiment.

[0215] Referring to FIG. **10A**, an electronic device (e.g., the electronic device **101** of FIG. **1** or the processor **120** of FIG. **1**) may acquire and group **1010** information related to a meeting schedule. According to an embodiment, when a content displayed on a display satisfies a configured condition, the electronic device may acquire and group information included in the content. For example, a first application (e.g., an Internet application, a booking agency application, and a restaurant's own application) may receive a user input of inputting the number of people and a specific place related to a meeting.

[0216] According to an embodiment, the electronic device may acquire (e.g., extract) a meeting-related content via a content (e.g., text or an image) included in an execution screen of the first application. For example, the electronic device may acquire a meeting place, meeting time information, and/or information on meeting attendees via a restaurant booking screen of the first application.

[0217] According to an embodiment, the electronic device may acquire restaurant-related information via an execution screen of a second application (e.g., the Internet application and an SNS application). For example, the electronic device may analyze a content (e.g., text or an image) included in the execution screen of the second application so as to acquire restaurant-related information, such as a main menu item and/or discount information of a restaurant.

[0218] According to an embodiment, the electronic device may acquire place-related information, such as a restaurant location and/or contact information, via a content (e.g., text or an image) included in an execution screen of a third application (e.g., the Internet application and a map application).

[0219] According to an embodiment, the electronic device may acquire the number of attendees and/or attendee name information via a content (e.g., text or an image) included in an execution screen of a fourth application (e.g., an email application and a message application).

[0220] According to an embodiment, the electronic device may group the acquired information into one group depending on a purpose. For example, the electronic device may group the acquired information into one schedule. According to an embodiment, the electronic device may perform grouping by further adding, to the information acquired via the content, information received from

an external server and information acquired by a user input.

[0221] According to an embodiment, the electronic device may classify the grouped information into the plurality of items. For example, the plurality of items may include a schedule (e.g., a meeting date/time) item, a place item, an attendee item, a parking lot information item, a recommended menu item, and/or a discount information item.

[0222] According to an embodiment, the electronic device may arrange **1020** information, based on priorities. For example, the electronic device may arrange the plurality of items obtained by classifying the grouped information, based on a configured first priority.

[0223] According to an embodiment, in a case of the plurality of meeting-related items, the first priority may be according to importance required for the progress of a meeting. For example, arrangement may be performed so that the schedule item, which is information required to attend a meeting, has the highest priority, and the meeting place item and the attendee item have subsequent priorities in sequence.

[0224] According to an embodiment, when the electronic device is able to perform grouping into one group (e.g., a schedule or a purchase list) via information acquired for a certain period (e.g., before September 12), the electronic device may classify the acquired information into the plurality of items and arrange the same according to the first priority, and may provide the plurality of arranged items **1030** to a user. According to an embodiment, each of the plurality of items **1030** may include a link for providing of summary information related to the item and/or detailed information related to the item. For example, the link may include an icon of an application that provides detailed information related to the item.

[0225] According to an embodiment, the electronic device may rearrange priorities over time or according to a change of information, which will be described below with reference to FIG. **10B** to FIG. **10D**.

[0226] FIG. **10B** is a diagram for illustrating electronic device operations of rearranging, based on priorities, the plurality of items and displaying the plurality of rearranged items, according to an embodiment. For example, FIG. **10B** illustrates a case in which the priorities of the plurality of items are changed over time.

[0227] Referring to FIG. **10B**, if a specific time point arrives before starting a schedule on a scheduled day, the electronic device may rearrange the plurality of items so that a parking lot information item **1040** has the highest priority, and may display a time required to arrive at the meeting place and a time to depart.

[0228] According to an embodiment, the electronic device may rearrange the plurality of items so that the schedule item previously having the highest priority and a place item **1041** having a subsequent priority have priorities lowered by one level.

[0229] FIG. **10C** is a diagram for illustrating electronic device operations of rearranging, based on priorities, the plurality of items and displaying the plurality of rearranged items, according to an embodiment. For example, FIG. **10C** illustrates a case in which the priorities of the plurality of items are changed over time.

[0230] Referring to FIG. **10C**, after checking departure and before arrival, the electronic device may rearrange the plurality of items so that a recommended menu item **1050** has the highest priority, and may display a recommended menu.

[0231] FIG. **10D** is a diagram for illustrating electronic device operations of rearranging, based on priorities, the plurality of items and displaying the plurality of rearranged items, according to an embodiment. For example, FIG. **10D** illustrates a case in which the priorities of the plurality of items are changed over time.

[0232] Referring to FIG. **10D**, the electronic device may rearrange the plurality of items so that a discount information item **1060** has the highest priority during the progress of the schedule, and may display information related to a discount card, a discount condition, and/or a discount rate.

[0233] FIG. **11A** is a diagram for illustrating electronic device operations of acquiring information,

arranging the plurality of items based on the acquired information according to priorities, and displaying the plurality of arranged items, according to an embodiment.

[0234] Referring to FIG. 11A, an electronic device (e.g., the electronic device **101** of FIG. 1 or the processor **120** of FIG. 1) may acquire and group **1110** information related to a meeting schedule. According to an embodiment, when a content displayed on a display satisfies a configured condition, the electronic device may acquire and group information included in the content. For example, a first application (e.g., a shopping mall application, an Internet application, and an SNS application) may receive a user input to retrieve some of products.

[0235] According to an embodiment, the electronic device may acquire retrieved product-related information via a content (e.g., text or an image) included in an execution screen of the first application. For example, the electronic device may acquire a product name, option information, and/or information on a product price via a product description screen of the first application.

[0236] According to an embodiment, the electronic device may receive a user input to retrieve some of products via an execution screen of a second application (e.g., the shopping mall application, the Internet application, and the SNS application), and may acquire product-related information by analyzing a content (e.g., text or an image) included in the execution screen of the second application.

[0237] According to an embodiment, when a user input to capture the content (e.g., text or an image) included in the execution screen of the second application is received, the electronic device may acquire information included in a captured image, as the product-related information.

[0238] According to an embodiment, the electronic device may acquire the product-related information via text included in an execution screen of a third application (e.g., the email application, the message application, and a document writing application).

[0239] According to an embodiment, the electronic device may group the acquired information into one group depending on a purpose. For example, the electronic device may group the acquired information into one schedule. According to an embodiment, the electronic device may perform grouping by further adding, to the information acquired via the content, information received from an external server and information acquired by a user input.

[0240] According to an embodiment, the electronic device may classify the grouped information into the plurality of items. For example, the plurality of items may include a purchase goods list item and/or a purchase place item. According to an embodiment, information on the purchase place item may be information on a place where multiple goods included in a purchase goods list can be purchased the most, and may include a location of an offline store or an online site address.

[0241] According to an embodiment, the electronic device may arrange **1120** information, based on priorities. For example, the electronic device may arrange the plurality of items obtained by classifying the grouped information, based on a configured first priority.

[0242] According to an embodiment, in a case of the plurality of purchase list-related items **1130** and **1131**, the first priority may be according to importance required for purchasing a product. For example, arrangement may be performed so that the place item **1130**, which is information for purchasing a product, has the highest priority, and the purchase goods list item **1131** has a priority in sequence.

[0243] According to an embodiment, the electronic device may arrange, based on the priorities, multiple products included in the purchase goods list item **1131**. For example, the electronic device may arrange the multiple products, based on a sequence in which the products are retrieved, or based on whether or not the products are daily use items.

[0244] According to an embodiment, when the electronic device is able to perform grouping into one group (e.g., a schedule or a purchase list) via information acquired for a certain period (e.g., before September 1), the electronic device may classify the acquired information into the plurality of items and arrange the same according to the first priority, and may provide the plurality of arranged items **1130** and **1131** to a user. According to an embodiment, at least a part of the plurality

of items **1130** and **1131** may include links for providing item-related summary information and/or item-related detailed information. For example, the link may include an icon of an application that provides the item-related detailed information.

[0245] According to an embodiment, the electronic device may rearrange the priorities over time or according to a change of information, which will be described below with reference to FIG. **11B** and FIG. **11C**.

[0246] FIG. **11B** is a diagram for illustrating electronic device operations of rearranging, based on priorities, the plurality of items and displaying the plurality of rearranged items, according to an embodiment. For example, FIG. **11B** illustrates a case in which the priorities of the plurality of items are changed over time and according to an information change.

[0247] Referring to FIG. **11B**, the electronic device may display purchased goods (e.g., milk and water) as being deleted from a purchase goods list **1141** so as to indicate that the purchase has been completed, and may rearrange the priority thereof to have a lower priority.

[0248] According to an embodiment, when a discount period related to goods (e.g., sneakers) required to be purchased arrives with the passage of time, the electronic device may change information of the place item **1140** to information on a place where the discount is performed. According to an embodiment, the electronic device may change the information of the place item **1140** to information on the place where the discount is performed, and may display a discount rate and/or discount period information.

[0249] FIG. **11C** is a diagram for illustrating electronic device operations of rearranging, based on priorities, the plurality of items and displaying the plurality of rearranged items, according to an embodiment. For example, FIG. **11C** illustrates a case in which the priorities of the plurality of items are changed over time and according to an information change.

[0250] Referring to FIG. **11C**, the electronic device may display purchased goods (e.g., milk, water, and sneakers) as being deleted from a purchase goods list **1152** so as to indicate that the purchase has been completed, and may rearrange the priority thereof to have a lower priority.

[0251] According to an embodiment, when a discount period related to goods required to be purchased (e.g., a cat tower) arrives with the passage of time, the electronic device may rearrange the plurality of items so that a discount period item **1150** and a discount place item **1151** have the highest priority. According to an embodiment, when a deadline (9/14) of the purchase list is near, the electronic device may change information of the discount place item **1151** to an offline place in order to recommend offline purchase rather than online purchase. According to an embodiment, in addition to rearrangement of the plurality of items, the electronic device may display discount rate information and optimal purchase time period information.

[0252] According to an embodiment, an electronic device may include a display, a memory, and at least one processor, wherein the at least one processor is configured to execute instructions to, if a content displayed on the display satisfies a configured condition, acquire information included in the content, arrange, based on a first priority, the plurality of items acquired based on the information included in the content so as to store the same in the memory, rearrange the plurality of items based on a second priority if a current time point satisfies the configured condition, and control the display to display the plurality of rearranged items.

[0253] According to an embodiment, if the content includes at least one of time information, place information, or an event name, the at least one processor may execute the instructions to determine that the configured condition is satisfied, and may acquire the information included in the content.

[0254] According to an embodiment, if a content satisfying the configured condition is displayed, the at least one processor may execute the instructions to display a user interface to store information related to the content in a partial area of the display, and may acquire the information included in the content when a user input to select the user interface is received.

[0255] According to an embodiment, the at least one processor may execute the instructions to acquire, if a first content included in an execution screen of a first application displayed on the

display satisfies the configured condition, information included in the first content, may acquire, if a second content included in an execution screen of a second application displayed on the display satisfies the configured condition, information included in the second content, and may acquire the plurality of items based on the information included in the first content and the information included in the second content.

[0256] According to an embodiment, the electronic device may further include a communication module, and the at least one processor may execute the instructions to acquire the plurality of items by further including information received from an external server via the communication module.

[0257] According to an embodiment, the at least one processor may execute the instructions to receive a user input to input information via the display, and may acquire the plurality of items by further including information input via the user input.

[0258] According to an embodiment, the at least one processor may execute the instructions to group the information included in the content into one schedule via analysis of at least one of an image or text included in the content, may classify the grouped information included in the content into the plurality of items, may rearrange the plurality of items based on the second priority if the current time point is before a first period of a time point related to the one schedule, and may rearrange the plurality of items based on a third priority if the current time point is before a second period that is shorter than the first period of the time point related to the one schedule.

[0259] According to an embodiment, the at least one processor may execute the instructions to control, when the plurality of items are rearranged, the display to display a notification of the rearrangement, and may control the display to display the plurality of rearranged items when a user input to select the notification is received.

[0260] According to an embodiment, when an event occurs in which information included in at least one of the plurality of items is changed, the at least one processor may execute the instructions to rearrange the plurality of items so that the item in which information has been changed has the highest priority.

[0261] According to an embodiment, the electronic device may further include a communication module, and the at least one processor may execute the instructions to identify that the event has occurred when the changed information for at least one of the plurality of items is received from an external server via the communication module.

[0262] According to an embodiment, a control method of an electronic device may include, if a content displayed on a display of the electronic device satisfies a configured condition, acquiring information included in the content, arranging, based on a first priority, the plurality of items acquired based on the information included in the content so as to store the same in a memory of the electronic device, rearranging the plurality of items, based on a second priority if a current time point satisfies the configured condition, and controlling the display to display the plurality of rearranged items.

[0263] According to an embodiment, the acquiring of the information included in the content may include, if the content includes at least one of time information, place information, or an event name, determining that the configured condition is satisfied, and acquiring the information included in the content.

[0264] According to an embodiment, the acquiring of the information included in the content may include, if a content satisfying the configured condition is displayed, displaying a user interface to store information related to the content in a partial area of the display, and acquiring the information included in the content when a user input to select the user interface is received.

[0265] According to an embodiment, the acquiring of the information included in the content may include acquiring, if a first content included in an execution screen of a first application displayed on the display satisfies the configured condition, information included in the first content, and acquiring, if a second content included in an execution screen of a second application displayed on the display satisfies the configured condition, information included in the second content, and the

arranging of the plurality of items based on the first priority and storing the same in a memory of the electronic device may include acquiring the plurality of items, based on the information included in the first content and the information included in the second content.

[0266] According to an embodiment, the arranging of the plurality of items based on the first priority and storing the same in a memory of the electronic device may include acquiring the plurality of items by further including information received from an external server.

[0267] According to an embodiment, the arranging of the plurality of items based on the first priority and storing the same in a memory of the electronic device may include receiving a user input to input information via the display, and acquiring the plurality of items by further including information input via the user input.

[0268] According to an embodiment, the arranging of the plurality of items based on the first priority and storing the same in a memory of the electronic device may include grouping the information included in the content into one schedule via analysis of at least one of an image or text included in the content, and classifying the grouped information included in the content into the plurality of items, and the rearranging of the plurality of items based on the second priority may include, if the current time point is before a first period of a time point related to the one schedule, rearranging the plurality of items based on the second priority, and if the current time point is before a second period that is shorter than the first period of the time point related to the one schedule, rearranging the plurality of items based on a third priority.

[0269] According to an embodiment, the method may further include, when the plurality of items are rearranged, controlling the display to display a notification of the rearrangement, and when a user input to select the notification is received, controlling the display to display the plurality of rearranged items.

[0270] According to an embodiment, the method may further include, when an event occurs in which information included in at least one of the plurality of items is changed, rearranging the plurality of items so that the item in which information has been changed has the highest priority.

[0271] According to an embodiment, the method may further include, when the changed information for at least one of the plurality of items is received from an external server, identifying that the event has occurred.

[0272] According to an embodiment, a non-transitory storage medium storing instructions configured to be executed by at least one processor of an electronic device to enable the electronic device to perform: based on a content displayed on a display of the electronic device satisfying a configured condition, acquiring information included in the content, arranging and storing in a memory of the electronic device, based on a first priority, a plurality of items acquired based on the information included in the content, based on a current time point satisfying the configured condition, rearranging the plurality of items based on a second priority, and controlling the display to display the plurality of rearranged items.

[0273] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0274] It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B”, “at least one of A and B”, “at least one of A or B”, “A, B, or C”, “at least one of A, B, and C”, and “at least one of A, B, or C”,

may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd”, or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with”, “coupled to”, “connected with”, or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0275] As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic”, “logic block”, “part”, or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0276] Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0277] According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0278] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components or operations may be omitted, or one or more other components or operations may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

Claims

- 1.** An electronic device comprising: a display; at least one processor including processing circuitry; and memory storing instructions that, when executed by the at least one processor individually or collectively, cause the electronic device to: obtain first information associated with a schedule from a plurality of applications; based on the first information, control the display to display a list of a plurality of items associated with the schedule; obtain second information associated with the schedule from a first application of the plurality of applications; identify, from among the plurality of items, a first item associated with the second information; and change an order of the first item in the list of the plurality of items.
- 2.** The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: obtain third information associated with the schedule from a second application of the plurality of applications; identify, from among the plurality of items, a second item associated with the third information; and delete the second item from the list of the plurality of items.
- 3.** The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: obtain fourth information associated with the schedule from a third application of the plurality of applications; generate a third item corresponding to the fourth information; and add the third item to the list of the plurality of items.
- 4.** The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: generate the plurality of items by classifying the first information into the plurality of items.
- 5.** The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: control the display to display an execution screen of each of the plurality of applications; and obtain the first information associated with the schedule from the execution screen.
- 6.** The electronic device of claim 5, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: display a user interface for obtaining the first information from the execution screen; and based on an input on the user interface being received, obtain the first information associated with the schedule from the execution screen.
- 7.** The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: based on the first item being identified, control the display to display a notification of rearrangement, and based on an input to select the notification being received, control the display to display the list of the plurality of items in which the order of the first item is changed.
- 8.** The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: based on the first item being identified, control the display to display the list of the plurality of items in which the order of the first item is changed so that the first item has highest priority.
- 9.** A control method of an electronic device, the method comprising: obtaining first information associated with a schedule from a plurality of applications; based on the first information, controlling a display of the electronic device to display a list of a plurality of items associated with the schedule; obtaining second information associated with the schedule from a first application of the plurality of applications; identifying, from among the plurality of items, a first item associated with the second information; and changing an order of the first item in the list of the plurality of items.
- 10.** The method of claim 9, further comprising: obtaining third information associated with the schedule from a second application of the plurality of applications; identifying from among the plurality of items, a second item associated with the third information; and deleting the second item

from the list of the plurality of items.

11. The method of claim 9, further comprising: obtaining fourth information associated with the schedule from a third application of the plurality of applications; generating a third item corresponding to the fourth information; and adding the third item to the list of the plurality of items.

12. The method of claim 9, further comprising: controlling the display to display an execution screen of each of the plurality of applications; and obtaining the first information associated with the schedule from the execution screen.

13. The method of claim 9, further comprising: based on the first item being identified, controlling the display to display a notification of rearrangement, and based on an input to select the notification being received, controlling the display to display the list of the plurality of items in which the order of the first item is changed.

14. The method of claim 9, further comprising: based on the first item being identified, controlling the display to display the list of the plurality of items in which the order of the first item is changed so that the first item has highest priority.

15. A non-transitory storage medium storing instructions that, when executed by at least one processor of an electronic device individually or collectively, cause the electronic device to perform a plurality of operations, the plurality of operations comprising: obtaining first information associated with a schedule from a plurality of applications; based on the first information, controlling a display of the electronic device to display a list of a plurality of items associated with the schedule; obtaining second information associated with the schedule from a first application of the plurality of applications; identifying, from among the plurality of items, a first item associated with the second information; and changing an order of the first item in the list of the plurality of items.

16. The non-transitory storage medium of claim 15, the plurality of operations further comprising: obtaining third information associated with the schedule from a second application of the plurality of applications; identifying from among the plurality of items, a second item associated with the third information; and deleting the second item from the list of the plurality of items.

17. The non-transitory storage medium of claim 15, the plurality of operations further comprising: obtaining fourth information associated with the schedule from a third application of the plurality of applications; generating a third item corresponding to the fourth information; and adding the third item to the list of the plurality of items.

18. The non-transitory storage medium of claim 15, the plurality of operations further comprising: controlling the display to display an execution screen of each of the plurality of applications; and obtaining the first information associated with the schedule from the execution screen.

19. The non-transitory storage medium of claim 15, the plurality of operations further comprising: based on the first item being identified, controlling the display to display a notification of rearrangement, and based on an input to select the notification being received, controlling the display to display the list of the plurality of items in which the order of the first item is changed.

20. The non-transitory storage medium of claim 15, the plurality of operations further comprising: based on the first item being identified, controlling the display to display the list of the plurality of items in which the order of the first item is changed so that the first item has highest priority.
